

Week 6 Task

Introduction to Wireless Communication

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Summary

This project uses Deep Reinforcement Learning to solve a common problem in wireless networks: finding the right balance for a transmitter's power level. If the power is set too low, the signal becomes weak and noisy, but setting it too high wastes energy unnecessarily. By building a simulated environment that accounts for user distance and natural signal fading, we trained a smart agent that automatically learns to adjust its transmit power step-by-step—using just enough energy to keep the connection clear without overshooting.